

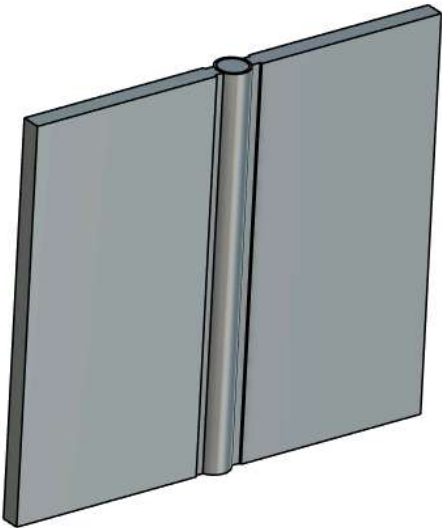


First-Sim_HingeV3



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Analyzed File	Hinge_V3 v3	Version	Autodesk Fusion (2703.1.11)
Creation Date	2026-05-29, 18:30:58	Author	Maxwell Berche
Summary	Static stress analysis of a steel door hinge assembly under 500N lateral load. Analysis identified the pin-to-plate interface as the critical failure zone with a minimum safety factor of 1.609, below the 2.00 engineering target. To meet safety requirements, the pin diameter should be increased or a higher-strength material considered.		

Report Properties

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Title	Studies



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Simulation Model 1

Study 1 - Static Stress

Study Properties

Study Properties	
Study Type	Static Stress
Last Modification Date	2026-05-29, 18:16:55

Study Setup

Settings		
General		
	Contact Tolerance	0.10 mm
	Remove Rigid Body Modes	No
Mesh		
	Average Element Size (% of model size) - Solids	10
	Scale Mesh Size Per Part	No
	Average Element Size (absolute value)	-
	Element Order	Parabolic
	Create Curved Mesh Elements	Yes
	Max. Turn Angle on Curves (Deg.)	60
	Max. Adjacent Mesh Size Ratio	1.5
	Max. Aspect Ratio	10
	Minimum Element Size (% of average size)	20
Adaptive Mesh Refinement		
	Number of Refinement Steps	0



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Results Convergence Tolerance (%)	20
Portion of Elements to Refine (%)	10
Results for Baseline Accuracy	von Mises Stress

Materials		
Steel		
Components		
	Component	Strength
	pin v1:1	Yield Strength
	plate1 v1:1	Yield Strength
	plate1 v1:2	Yield Strength
Properties		
	Density	7.850E-06 kg / mm^3
	Young's Modulus	210000.00 MPa
	Poisson's Ratio	0.30
	Yield Strength	207.00 MPa
	Ultimate Tensile Strength	345.00 MPa
	Thermal Conductivity	0.056 W / (mm C)
	Thermal Expansion Coefficient	1.200E-05 / C
	Specific Heat	480.00 J / (kg C)

Contacts				
Contact Type	Contact Set	Penetration Type	Bodies	Entities
Bonded				
	Bonded1	Symmetric	[Body1 plate1 v1:2 Body1 pin v1:1]	[Face 0 Face 0]
	Bonded2	Symmetric	[Body1 plate1 v1:1 Body1 plate1 v1:2]	[Face 3 Face 0]
	Bonded3	Symmetric	[Body1 plate1 v1:1 Body1 pin v1:1]	[Face 0 Face 0]
	Bonded4	Symmetric	[Body1 plate1 v1:1 Body1 plate1 v1:2]	[Face 0 Face 3]

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Type	Solids



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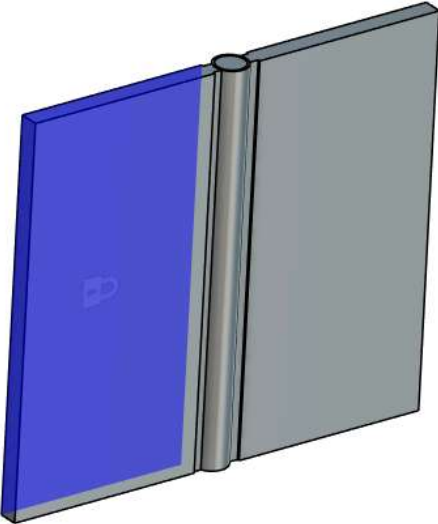
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Nodes	20842
Elements	11165

Load Case1

Constraints	
Fixed1	
	
Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

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Force3	



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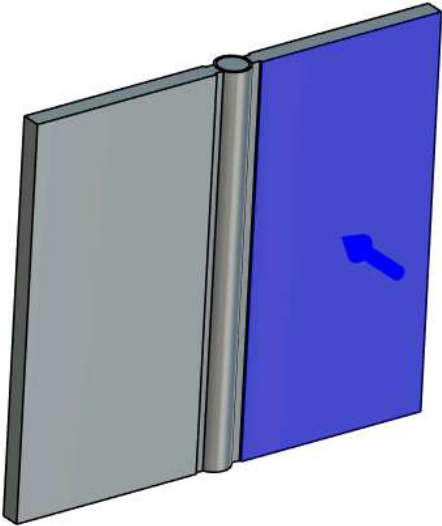
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Type	Force
Magnitude	500.00 N
X Value	-500.00 N
Y Value	0.00 N
Z Value	0.00 N
Force Per Entity	No

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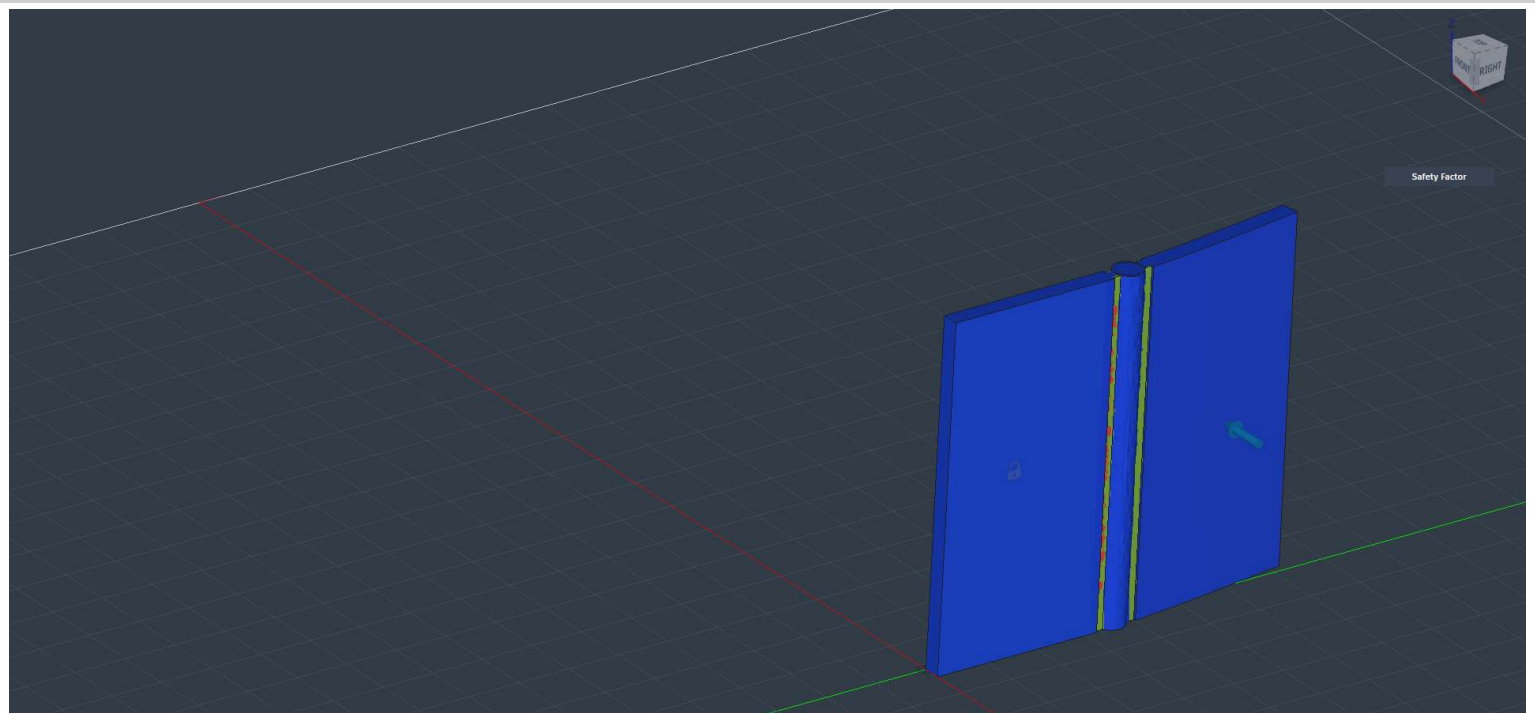
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● With the analysis criteria in this study setup, the design is expected to bend permanently or break.

Next Steps

Below Safety Factor Target

Validate your study setup

- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Strengthen weak areas

- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Generate alternative designs



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- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Meets Safety Factor Target

Validate your study setup

- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Above Safety Factor Limit

Validate your study setup

- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Lightweight the design

- Set a Safety Factor upper limit to find where material can be removed
- Remove material from areas that exceed the Safety Factor upper limit

Generate alternative designs

- In the Generative Design workspace, create design modifications
- Remove unnecessary material using a Shape Optimization study

Load Case1 - Results



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Result Summary			
Result type	Sub-result	Minimum	Maximum
Safety Factor			
	Safety Factor	1.609	127,769.594
Stress			
	Von Mises	0.002 MPa	128.672 MPa
	1st Principal	-56.448 MPa	143.584 MPa
	3rd Principal	-165.745 MPa	28.09 MPa
	Normal XX	-91.177 MPa	31.684 MPa
	Normal YY	-159.506 MPa	143.072 MPa
	Normal ZZ	-61.732 MPa	46.484 MPa
	Shear XY	-65.139 MPa	32.023 MPa
	Shear YZ	-17.021 MPa	12.793 MPa
	Shear XZ	-19.132 MPa	26.777 MPa
Displacement			
	Total	0 mm	0.124 mm
	X	-0.123 mm	1.149E-04 mm
	Y	-0.006 mm	0.007 mm
	Z	-4.515E-04 mm	4.451E-04 mm
Reaction Force			
	Total	0 N	111.771 N
	X	-25.643 N	80.481 N
	Y	-26.263 N	77.561 N
	Z	-6.157 N	19.765 N
Strain			
	Equivalent	9.382E-09	9.633E-04
	1st Principal	-5.027E-06	7.832E-04
	3rd Principal	-9.906E-04	1.811E-06



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	Normal XX	-3.086E-04	2.346E-04
	Normal YY	-6.398E-04	5.938E-04
	Normal ZZ	-1.117E-04	8.621E-05
	Shear XY	-8.065E-04	3.965E-04
	Shear YZ	-2.107E-04	1.584E-04
	Shear XZ	-2.369E-04	3.315E-04
Contact Pressure			
	Total	0 MPa	94.188 MPa
	X	-13.555 MPa	12.10 MPa
	Y	-82.126 MPa	93.695 MPa
	Z	-9.69 MPa	7.231 MPa
Contact Force			
	Total	0 N	154.136 N
	X	-61.257 N	67.218 N
	Y	-135.682 N	140.37 N
	Z	-15.835 N	18.672 N

Safety Factor
Safety Factor



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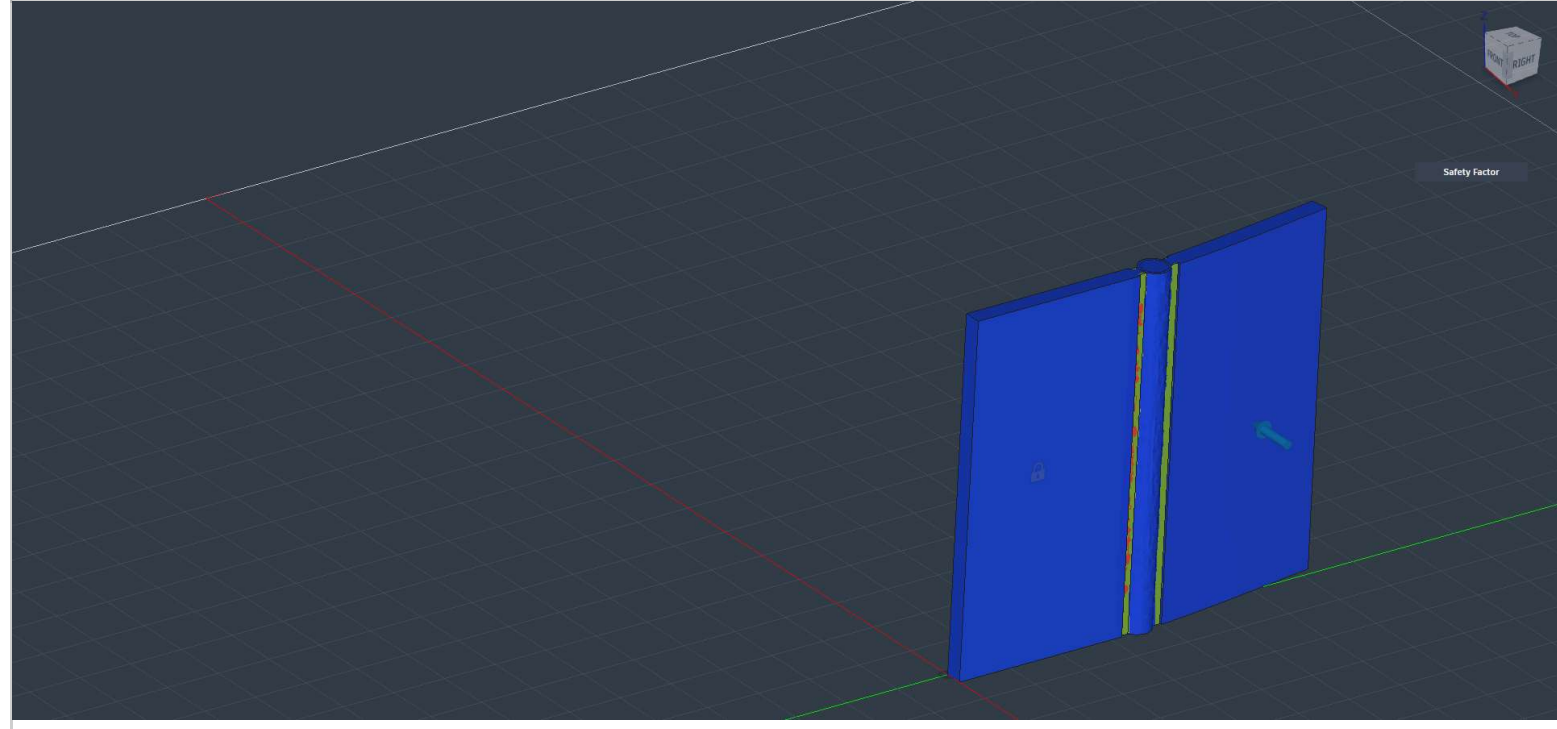
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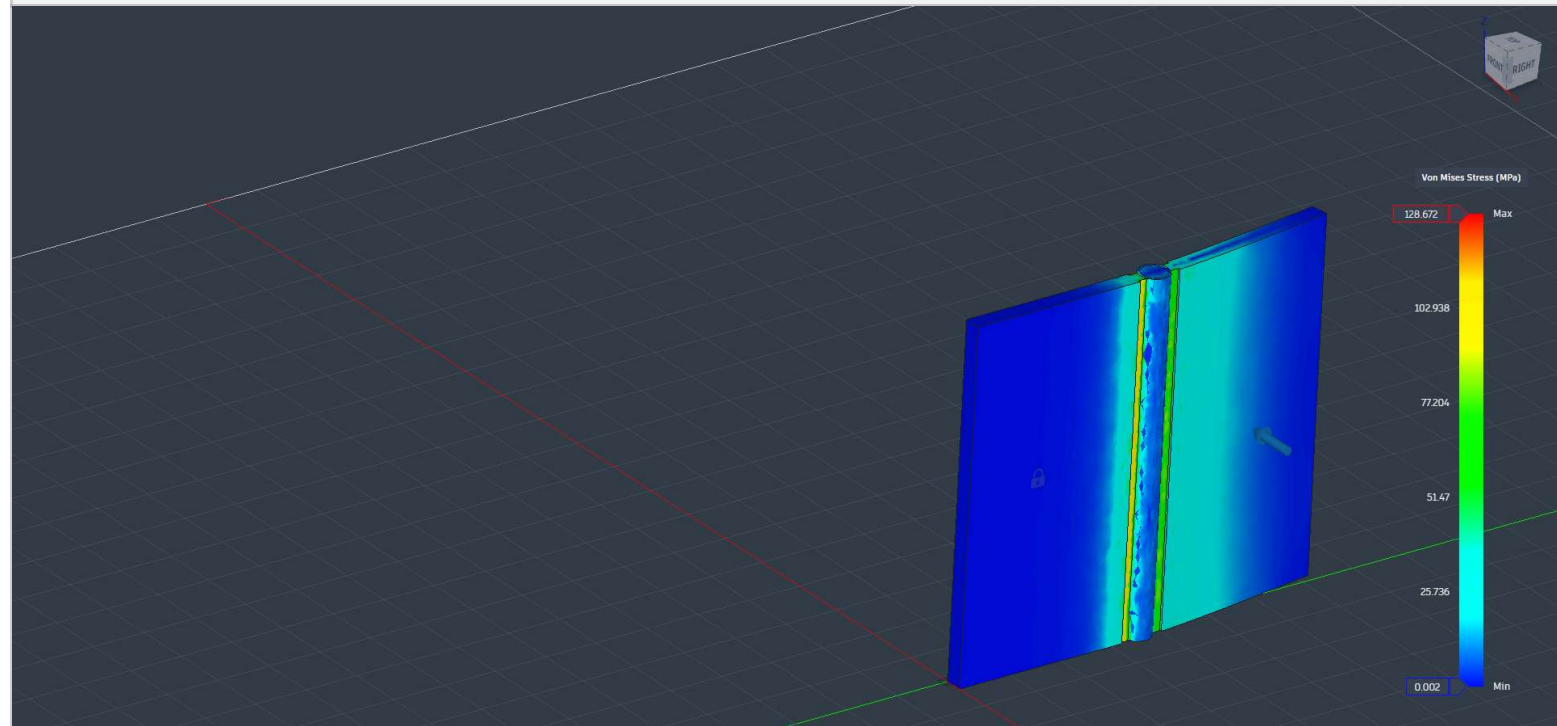
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Stress

Von Mises





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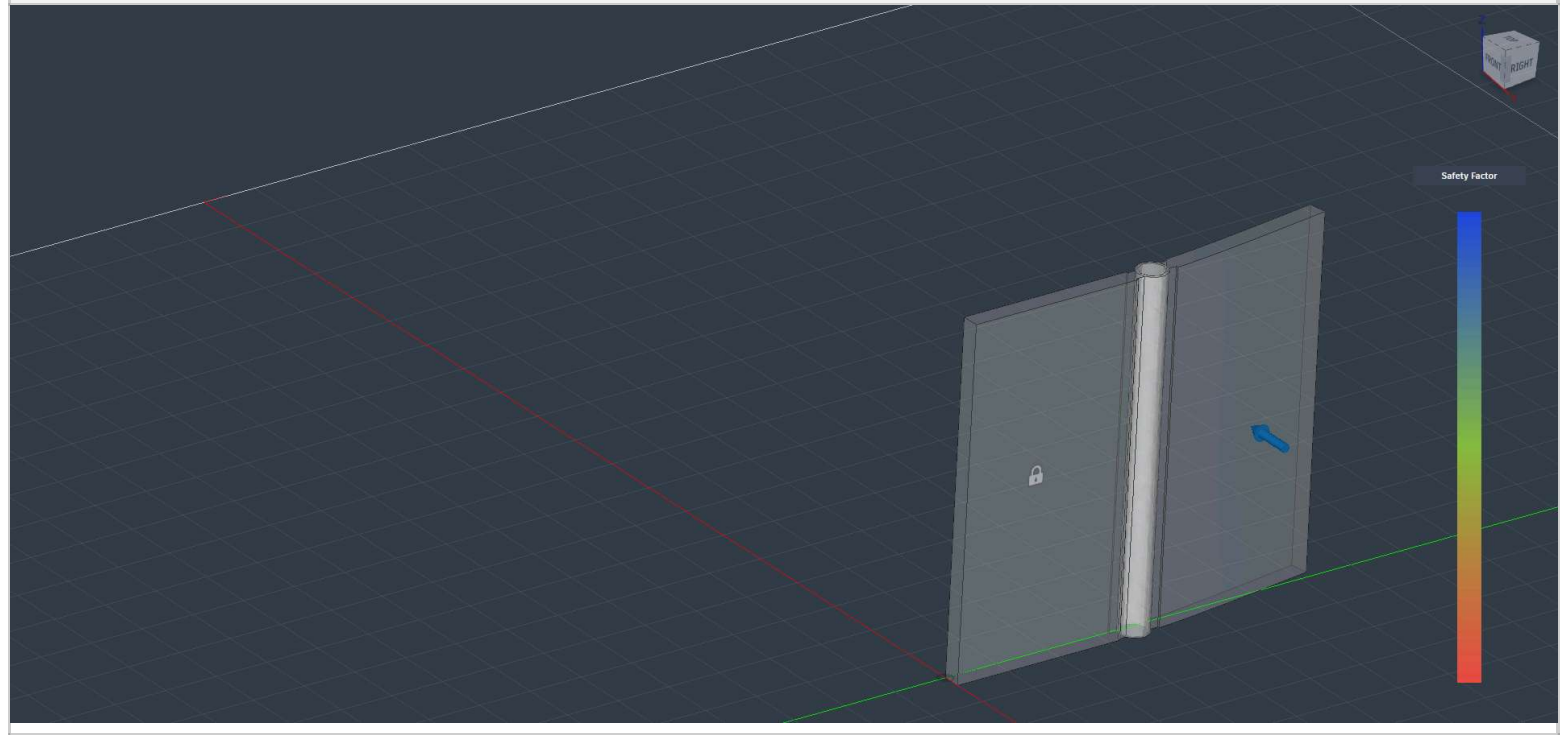
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1st Principal



Displacement

Total



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